

Introduction to Aerospace

Assessment of Aircraft Performance and Stability through Wind-Tunnel and Flight testing

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Answer the following questions within the space provided using 11pt Arial. Make sure tables and figures are properly formatted and readable. Marks will be deducted for bad formatting.

Q1. Plot your results for aircraft Lift Coefficient (C_L) vs Angle of Attack (α) obtained from both the wind tunnel and flight tests. Concentrate on the lift curve slope (a), zero-lift angle of attack (α_0) and max lift coef ($C_{L_{max}}$). Comment on the impact of the tailplane on the lift curve. How do the tunnel and flight test results compare? What might be the cause of any discrepancies?

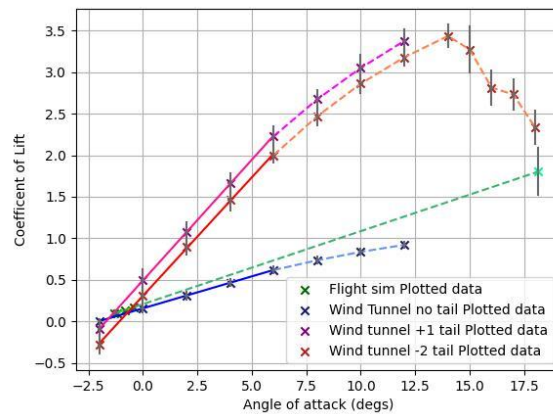


Fig 1: Lift coef vs Angle of attack

Table 1: Key statistics for each test.

Test	Lift curve slope (rad^{-1})	Zero-lift AoA (deg)	Max (recorded) lift coef
Wind tunnel -2 tailplane	0.285	-1.077	3.43354624
Wind tunnel +1 tailplane	0.291	-1.694	3.3756906
Wind tunnel no tailplane	0.078	-1.989	0.921795
Flight sim	0.088	-2.37	1.8034119

It is important to note that for the wind tunnel with no tail and tail at +2 deg the max lift coefficient may be slightly higher than stated as data at sufficiently high angle of attacks to properly observe stall was not recorded. Furthermore, the wool tufts might decrease the AoA at which separation occurs as they alter the wing's surface. The data collected for the flight simulation and the wind tunnel without a tail are very similar with only a 0.01 rad^{-1} difference in gradient. The error bars overlap when the data is plotted, due to the extension out to the max recorded lift coefficient the gradients seem to vary further, this could be due to the uncertainty of the precise moment at which stall started. The wind tunnel at -2 deg and +1 deg are also very similar, and overlap error bars at all points, they also both have much greater lift curve slopes. This therefore implies the tailplane increases the lift curve slope however they also have lower zero lift angle of attacks compared to the flight sim and the wind tunnel without a tail.

(25% of marks)

Q2. Plot graphs of the aircraft drag polar (C_D vs C_L) and the variation of aerodynamic efficiency with lift coef (L/D vs C_L), as estimated through wind tunnel and flight testing. How do the results from the two sets of tests compare? What might the cause of any differences be?

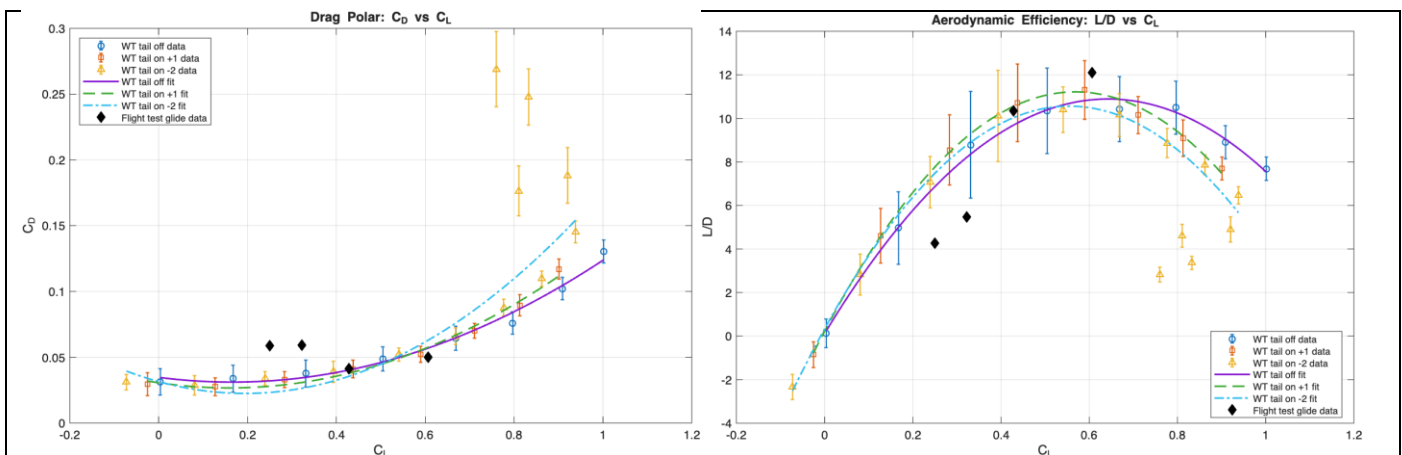


Figure 3 - Drag Polar: C_D vs C_L (left)

Figure 4 – Aerodynamic Efficiency: L/D vs C_L (right)

The drag polar shows the expected parabolic increase of C_D with C_L , while the L/D plot shows a clear peak at approximately $C_L = 0.5 - 0.6$, corresponding to maximum aerodynamic efficiency.

The wind tunnel data is smoother and follows a more consistent trend, whereas the flight-test data shows greater scatter. The flight-test points are broadly similar in trend but do not exactly overlap the tunnel results, with some points giving higher L/D near the efficient operating range. The tail-on -2° wind tunnel case shows larger scatter and higher drag at high C_L , suggesting flow separation or stall effects.

The differences are likely caused by Reynolds number differences between the scaled wind-tunnel model and the full-scale aircraft, wind tunnel wall blockage and support interference, and uncertainty in the flight glide test, especially maintaining constant IAS and timing the descent accurately.

(15% of marks)

Q3. From your data presented in Q2, predict the aircraft zero-lift drag coef. (C_{D_0}) and Oswald efficiency factor (e) given for each test and configuration. How does your aircraft's polar compare to the theoretical parabolic drag polar we have discussed in class? Why might differences arise.

	WT tail off	WT tail on+1	WT tail on+2
Oswald efficiency factor	0.574	0.485	0.530
zero-lift drag coefficient	0.028	0.032	0.030

Table 2: Oswald Efficiency Factor and Zero-Lift Drag Coefficient for each test

The experiment data is approximately valid for the theoretical parabolic model in operating range.

Oswald efficiency factors are around 0.5 which is half value with respect to the ideal value of $e=1$, causing twice the induced drag.

Theoretically, we use inviscid flow but in experimental case, viscosity exists. Turbulent flow would cause skin friction at wall. Adverse pressure gradient will lead to thickening of boundary layer which could eventually lead to separation. Then higher drag occurs

(15% of marks)

Q4. From your flight tests, plot the tailplane deflection required for trim (i_H) vs lift coef (C_L) for each of the different centre of gravity positions. Hence also plot the variation of the gradient di_H/dC_L with the centre of gravity position x_{cg} . What is the aircraft's neutral point position (x_{np}), effective tailplane lift curve slope ($\eta_H a_H$) and zero-lift pitching moment coef. (C_{M_0}). Comment on the level of uncertainty in your estimates.

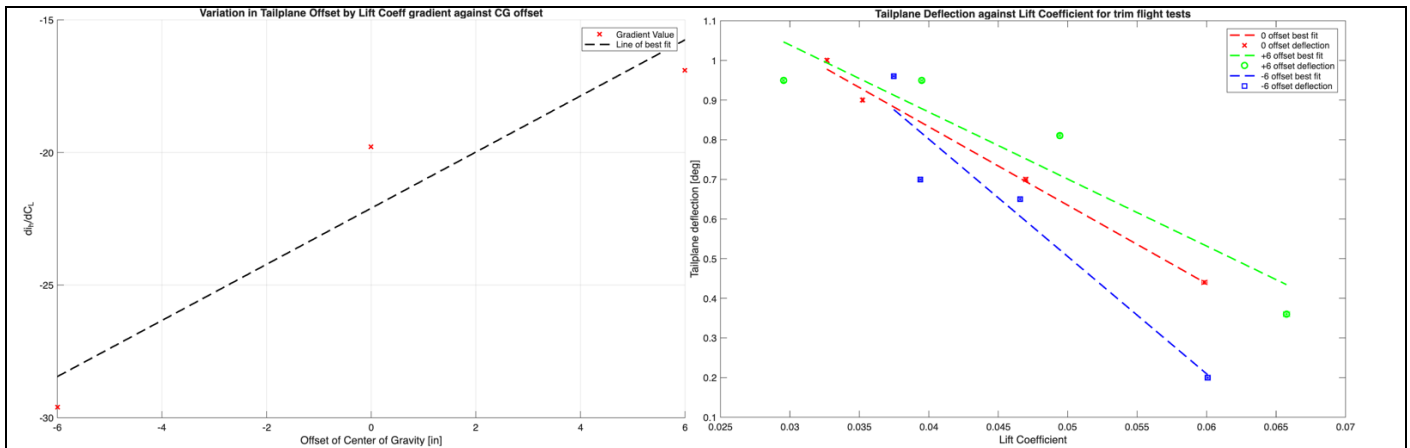


Figure 4 (left) – Rate of change of Tailplane offset with respect to Lift Coefficient against CG offset

Figure 5 (right) – Tailplane Deflection against Lift Coefficient for Trim Flight tests

Effective Tailplane Lift Curve Slope = 1.4992

Neutral Point = 20.8910in from CG offset 0in

Zero-Lift Pitching Moment Coefficient = 1.6246

Uncertainties arose in precision of data collection instruments, as well as more significant human error during use of flight simulator. The numerical deviation in values due to this was not recorded, so error bars are formed with respect to the precision with which values used in calculation were provided. This ultimately has led to very small error bars, which is not fully representative of the range of values we have recorded.

(15% of marks)

Q5. Comment on the appropriateness of any assumptions made in deriving the equation

$$(i_H)_e = \frac{C_{M_0} - K_n(C_L)_e}{\eta_H \bar{V}_H a_H}$$

which your prior analysis of flight tests data has relied upon. To what extent do these assumptions reflect the behaviour of the aircraft based on your measurements and how might they have impacted your estimates in Q4?

In the derivation, we assumed that altitude perturbations were negligible, implying that the aircraft was stable w.r.t these changes. Human error likely accounts for a reflection of more significant perturbations than expected by theory. Small angle approximations as well as the linear relationship between angle of attack and lift coefficient, may have been violated for similar reasons. We assumed a constant tail efficiency as well as lift curve slope - not accounting for effects of downwash, which would vary our results significantly. Estimates in Q4 were performed by extrapolating our data beyond points at which we recorded values. This too would be affected by human error. The numerical precision of provided values and recorded values is insignificant. The largest error source is thus in the recording of the tail-plane deflection, affecting pitching moment calculation (found using y intercept), as well as the lift curve slope (gradient).

(10% of marks)

Q6. Using your wind tunnel test data, plot a graph of pitching moment coef. (C_M) vs lift coef. (C_L) for the three cases tested. Hence, using the methods detailed in the laboratory handout determine the aircraft's neutral point position (x_{np}), effective tailplane lift curve slope ($\eta_H a_H$), the downwash angle derivative $d\varepsilon/d\alpha$, and the zero-lift pitching moment coef. (C_{M_0}). Comment on the level of uncertainty in your estimates. Why might the values of C_{M_0} estimated through flight testing and wind tunnel testing differ?

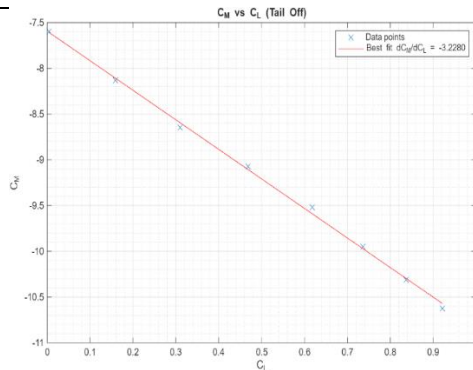


Fig 6 – C_m vs C_L Tailplane off

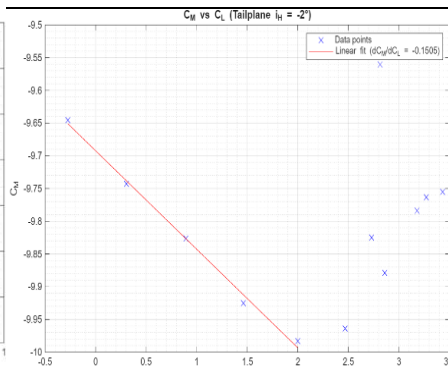


Fig 7 – C_m vs C_L Tailplane -2 deg

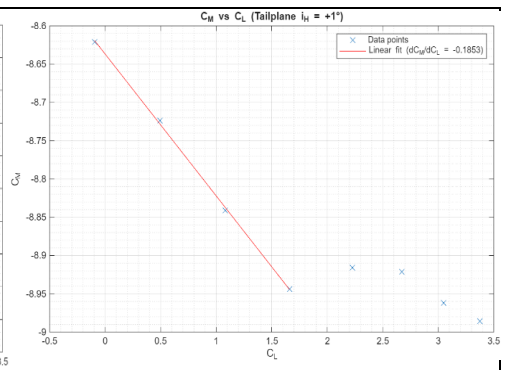


Fig 8 – C_m vs C_L Tailplane 1 deg

Table 3 – Key features of the model for each configuration

Tail Plane	K_n	x_{np} aft (m)	$\eta_H a_H$ (deg ⁻¹)	$d\varepsilon/d\alpha$	C_{M_0}
off	3.2280	0.9360	N/A	0.96	-7.599
+1	0.1853	0.4553	0.2480	0.96	-8.639
-2	0.1505	0.4498	0.2480	0.96	-8.699

C_{M_0} is likely different in both cases as in the wind tunnel the effects of the line at which thrust acts along isn't included which will have an effect in the flight sim as the line likely lies out of the centre of gravity of the aircraft. Additionally there isn't dynamically similarity as Reynolds number in both cases are different which means boundary layer behaviour is different in both.

(15% of marks)

Q7. Based on your plot in Q6 and your observations of the wool tufts, comment on the stalling behaviour of the aircraft. Does the aircraft exhibit a stable pitch break at stall? Is that consistent with your experience flight testing the aircraft's stalling behaviour?

During stall, separation of wool tufts started at the centre of the wing before moving gradually towards the wing tip and root, and vortices were generally the most pronounced at the centre of the wing. Tailplane wool tufts showed that the tailplane stalled at higher angles of attack than the main wing, uniformly along its trailing edge. For tail off testing, the constant $dC_M/dC_L < 0$ shows no pitch break at stall, following the same negative pitching moment (nose down). At $i_H = +1^\circ$, just after stalling there is a small increase in C_M followed by a trend of points that suggest a shallower $dC_M/dC_L < 0$ and thus a weak pitch break. After stalling, $i_H = -2^\circ$ shows a larger increase in C_M before it peaks, showing a stronger, more evident pitch break. Both show stable pitch break which causes the aircraft to naturally point nose down after stalling. This is consistent with the flight simulator aircraft's stalling behaviour, where at the onset of stall there is a subtle decrease to nose down pitching moment, followed by a gentler nose down pitching moment than present prior to stall, requiring more pilot input to recover the stall.

(5% of marks)